FORMULARIO TRIGONOMÉTRICO ALGEBRA Y TRIGONOMETRÍA MÓDULO 2

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Valores exactos de:

α	$\pi/6$	$\pi/4$	$\pi/3$
sen	$1/2$	$\sqrt{2}/2$	$\sqrt{3}/2$
cos	$\sqrt{3}/2$	$\sqrt{2}/2$	$1/2$
tan	$1/\sqrt{3}$	1	$\sqrt{3}$

Fórmulas suma y resta de ángulos

$$\begin{aligned}
\operatorname{sen}(\alpha \pm \beta) &= \operatorname{sen}\alpha\cos\beta \pm \cos\alpha\operatorname{sen}\beta\\
\cos(\alpha \pm \beta) &= \cos\alpha\cos\beta \mp \operatorname{sen}\alpha\operatorname{sen}\beta\\
\tan(\alpha \pm \beta) &= \frac{\tan\alpha \pm \tan\beta}{1 \mp \tan\alpha\tan\beta}
\end{aligned}$$

Fórmulas reducción primer cuadrante

$$\begin{aligned}
\operatorname{sen}(\pi \pm \alpha) &= \mp\operatorname{sen}\alpha; & \operatorname{sen}(2\pi - \alpha) &= -\operatorname{sen}\alpha\\
\cos(\pi \pm \alpha) &= -\cos\alpha; & \cos(2\pi - \alpha) &= \cos\alpha\\
\tan(\pi \pm \alpha) &= \pm\tan\alpha; & \tan(2\pi - \alpha) &= -\tan\alpha
\end{aligned}$$

Valores para ángulos cuadrantales

	0	$\pi/2$	π	$3\pi/2$	2π
sen	0	1	0	-1	0
cos	1	0	-1	0	1
tan	0	$\nexists$	0	$\nexists$	0
csc	$\nexists$	1	$\nexists$	-1	$\nexists$
sec	1	$\nexists$	-1	$\nexists$	1
cot	$\nexists$	0	$\nexists$	0	$\nexists$

Fórmulas ángulos doble

$$\begin{aligned}
\operatorname{sen}(2\alpha) &= 2\operatorname{sen}\alpha\cos\alpha\\
cos(2\alpha) = 2\cos^2\alpha - 1 &= 1 - 2\operatorname{sen}^2\alpha\\
\tan(2\alpha) &= \frac{2\tan\alpha}{1-\tan^2\alpha}
\end{aligned}$$

Fómulas reducción de cofunciones

$$\begin{aligned}
\operatorname{sen}(\tfrac{\pi}{2} - \alpha) &= \cos\alpha; & \cos(\tfrac{\pi}{2} - \alpha) &= \operatorname{sen}\alpha\\
\sec(\tfrac{\pi}{2} - \alpha) &= \csc\alpha; & \csc(\tfrac{\pi}{2} - \alpha) &= \sec\alpha\\
\tan(\tfrac{\pi}{2} - \alpha) &= \cot\alpha; & \cot(\tfrac{\pi}{2} - \alpha) &= \tan\alpha
\end{aligned}$$

Signo funciones Circulares

	I	*II*	*III*	*IV*
sen	+	+	−	−
cos	+	−	−	+
tan	+	−	+	−

Fórmulas ángulos medios

$$\begin{aligned}
\operatorname{sen}\frac{\alpha}{2} &= \pm\sqrt{\frac{1-\cos\alpha}{2}}\\
\cos\frac{\alpha}{2} &= \pm\sqrt{\frac{1+\cos\alpha}{2}}\\
\tan\frac{\alpha}{2} &= \pm\sqrt{\frac{1-\cos\alpha}{1+\cos\alpha}} = \frac{\operatorname{sen}\alpha}{1+\cos\alpha}
\end{aligned}$$

Fómulas reducción de cofunciones para $\pi/2 + \alpha$

$$\begin{aligned}
\operatorname{sen}(\tfrac{\pi}{2} + \alpha) &= \cos\alpha; & \cos(\tfrac{\pi}{2} + \alpha) &= -\operatorname{sen}\alpha\\
\sec(\tfrac{\pi}{2} + \alpha) &= -\csc\alpha; & \csc(\tfrac{\pi}{2} + \alpha) &= \sec\alpha\\
\tan(\tfrac{\pi}{2} + \alpha) &= -\cot\alpha; & \cot(\tfrac{\pi}{2} + \alpha) &= -\tan\alpha
\end{aligned}$$

Identidades básicas

$$\begin{aligned}
\operatorname{sen}\alpha\cdot\csc\alpha &= 1\\
\cos\alpha\cdot\sec\alpha &= 1\\
\tan\alpha\cdot\cot\alpha &= 1
\end{aligned}$$

Funciones circulares pares e impares

$$\begin{aligned}
\operatorname{sen}(-\alpha) &= -\operatorname{sen}\alpha\\
\cos(-\alpha) &= \cos\alpha\\
\tan(-\alpha) &= -\tan\alpha
\end{aligned}$$

Teorema de los Senos

$$\frac{\operatorname{sen}\alpha}{a} = \frac{\operatorname{sen}\beta}{b} = \frac{\operatorname{sen}\gamma}{c}$$

Identidades fundamentales

$$\begin{aligned}
\operatorname{sen}^2\alpha + \cos^2\alpha &= 1\\
\tan^2\alpha + 1 &= \sec^2\alpha\\
\cot^2\alpha + 1 &= \csc^2\alpha
\end{aligned}$$

Período de funciones circulares

$$\begin{aligned}
\operatorname{sen}(\theta + 2k\pi) &= \operatorname{sen}\theta, & k &\in \mathbb{Z}\\
\cos(\theta + 2k\pi) &= \cos\theta, & k &\in \mathbb{Z}\\
\tan(\theta + k\pi) &= \tan\theta, & k &\in \mathbb{Z}
\end{aligned}$$

Teorema de los Cosenos

$$\begin{aligned}
a^2 &= b^2 + c^2 - 2bc\cos\alpha\\
b^2 &= a^2 + c^2 - 2ac\cos\beta\\
c^2 &= a^2 + b^2 - 2ab\cos\gamma
\end{aligned}$$